Ayush Baghel

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Education

Madhav Institute of Technology and Science (MITS) Gwalior, India
Bachelor of Technology — Information Technology *Sep 2024 – Apr 2028*

- Relevant Coursework: Data Structures & Algorithms, DBMS, Object-Oriented Programming, Computer Networks, Operating Systems

Technical Skills

Languages: C++, Python, JavaScript, HTML, CSS, SQL
Libraries & Frameworks: NumPy, Pandas, Matplotlib, Seaborn, scikit-learn, Flask
Databases: PostgreSQL, MongoDB
Data & BI Tools: Microsoft Power BI, Microsoft Excel (Advanced), Jupyter Notebook
Machine Learning: Supervised Learning, Random Forest, Decision Trees, Regression, RMSE, R^2
Developer Tools: Git, GitHub, VS Code, REST APIs

Projects

Amazon Sales Data Analysis | *Python, Pandas, Matplotlib, Power BI, Jupyter Notebook* | repo

- Performed end-to-end EDA on a large Amazon sales dataset, uncovering revenue trends, top-performing product categories, and regional distribution patterns using Pandas and Matplotlib in Jupyter Notebook.
- Built an interactive Power BI dashboard with 8+ visualizations covering monthly sales trends, category-level KPIs, and order-wise revenue, enabling data-driven business decision-making.

Online Secure Examination & Result Management System | *Python, Flask, PostgreSQL* | repo

- Architected a full-stack web app supporting 3,000+ students with RBAC across student, faculty, and admin roles; designed a normalized DB schema for concurrent sessions with SQL injection and CSRF mitigations.
- Automated grade computation and report generation, enabling near-instant result publishing post-examination.

Code Plagiarism Detection System | *Python, Flask, REST API, HTML/CSS* | repo

- Engineered a plagiarism detector using a 5-metric weighted scoring model: text similarity (35%), token analysis (20%), line-level comparison (20%), variable naming patterns (15%), and control flow structure (10%).

Car Price Prediction using Machine Learning | *scikit-learn, Pandas, Random Forest, Decision Tree* | *In Progress*

- Developing a supervised ML pipeline on a 300+ record, 9-feature Kaggle automotive dataset to predict used car resale prices using Random Forest and Decision Tree regressors.
- Conducting comparative model evaluation (RMSE, R^2) across regressors; preprocessing pipeline covers feature engineering, encoding, imputation, and train-test splitting via scikit-learn.

Leadership & Affiliations

USG — Internal Affairs, DebSoc (MITS MUN Club) 2024 – Present
Madhav Institute of Technology and Science, Gwalior

- Spearheaded internal operations and successfully planned and executed a 3-committee MUN conference with 150+ participants; managed delegate onboarding, logistics, and cross-team coordination.

Member — Google Developer Group (GDG) on Campus 2024 – Present
MITS Student Chapter, Gwalior

- Active participant in GDG events, workshops, and study jams covering Google Cloud, Android, and ML technologies.

Additional

Languages: English (Professional), Hindi (Native)
Interests: Machine Learning, Data Analytics, Software Development, Open-Source Contribution